

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 (Suggested maximum time: 5 minutes)

(Suggested maximum time: 5 minutes)

Paulo picked the following numbers from a box of number cards.

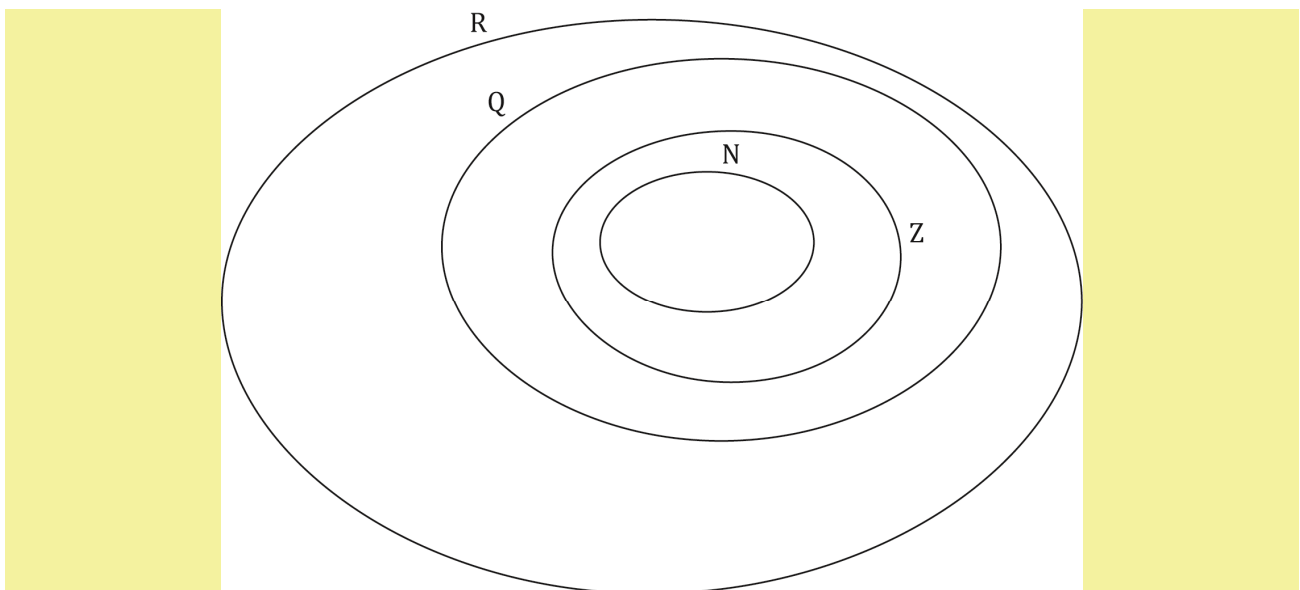
4	$\frac{23}{6}$	3.02	π	$(2.3)^{\frac{3}{2}}$	$2\sqrt{3}$
---	----------------	------	-------	-----------------------	-------------

- (a) Write these numbers in **ascending** order.

[illegible]

- (b)** The diagram below represents the sets:
Natural Numbers $\mathbb{N}$, Integers $\mathbb{Z}$, Rational Numbers $\mathbb{Q}$ and Real Numbers $\mathbb{R}$.

- (i) Insert each of the numbers listed above in the correct place on the diagram.

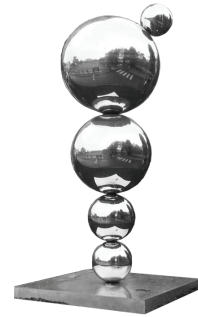


- (ii) The sets are in order as you move outwards from the centre set N . Explain why this is possible.

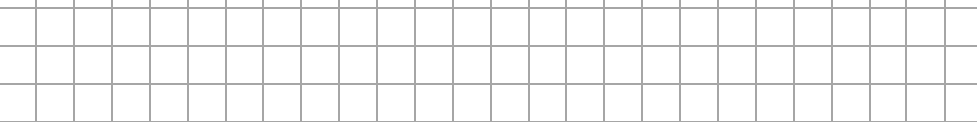
[illegible]

Question 2 (Suggested maximum time: 10 minutes)

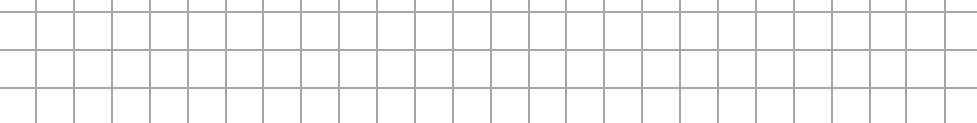
The Apples and Atoms sculpture pictured is situated in the grounds of Trinity College Dublin. The sculpture is made of five spheres of different sizes.



- (a) The smallest sphere has a diameter of 0.4 m.
Work out the **volume** of the smallest sphere.
Give your answer in m^3 , correct to two decimal places.

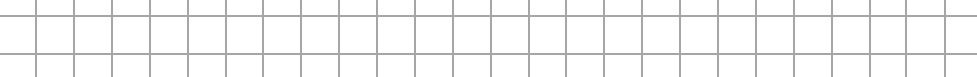


- (b)** The largest sphere has a volume of $4.5\pi \text{ m}^3$.
Work out the **radius** of the largest sphere.



- (c) The radii of each of spheres form a linear sequence. Fill in the radii in the table below.

Sphere	Smallest	2nd Sphere	3rd Sphere	4th Sphere	Largest
Radius (m)					

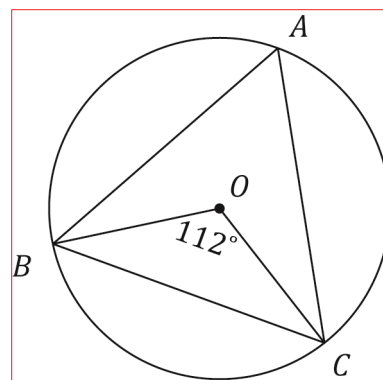
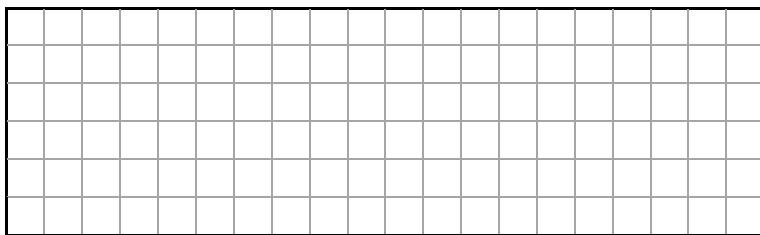


Question 3

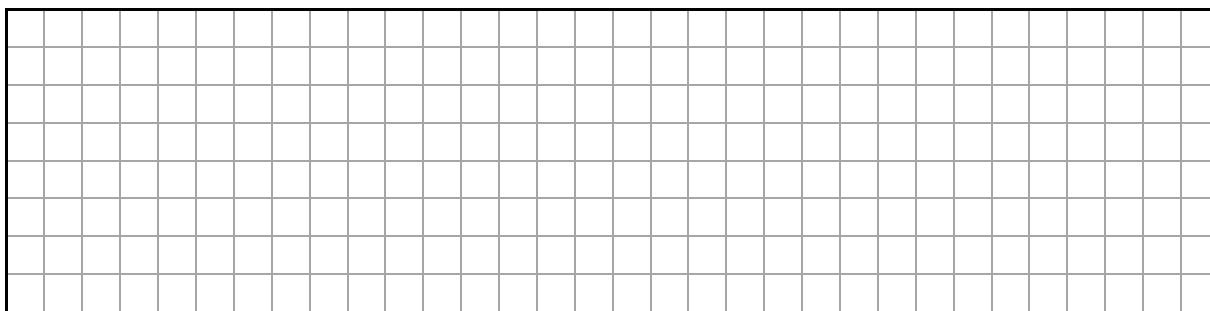
(Suggested maximum time: 10 minutes)

- (a) A, B and C are points on a circle with centre O .
 $|\angle COB| = 112^\circ$ and $|AB| = |AC|$, as shown.

- (i) Find $|\angle CAB|$.

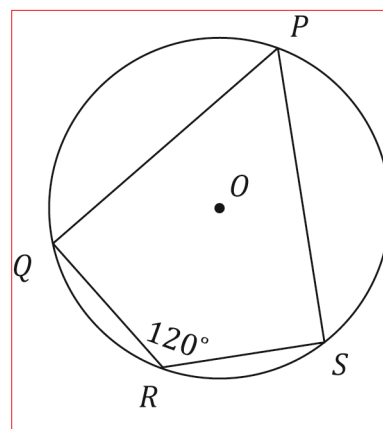
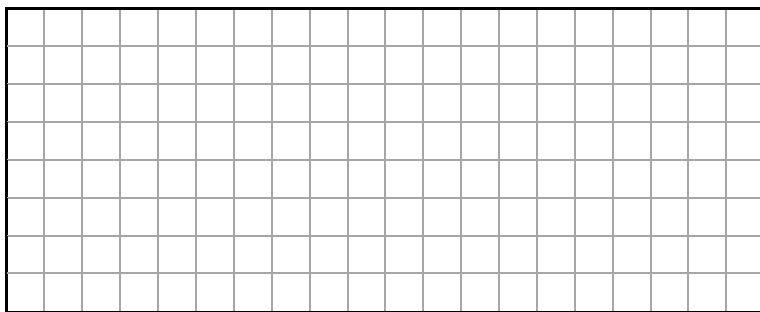


- (ii) Find $|\angle ABO|$.

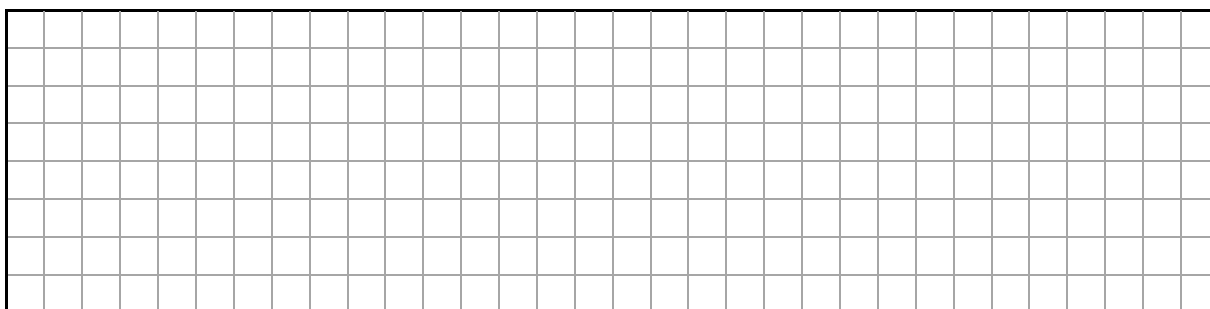


- (b) P, Q, R and S are points on another circle.
 $|\angle QRS| = 120^\circ$, as shown.

- (i) Find $|\angle SPQ|$.

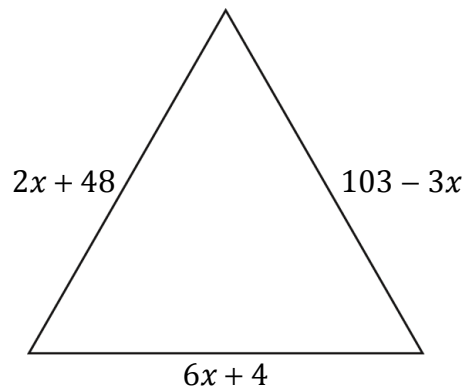


- (ii) The line PQ is perpendicular to QR .
 Explain why PR is a diameter of the circle.

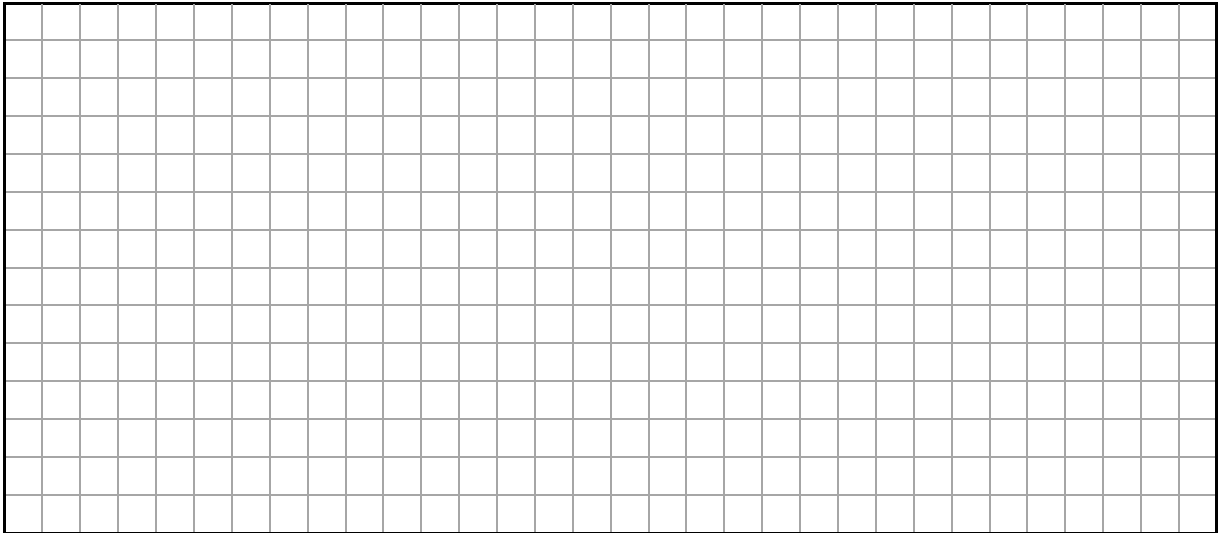


Question 4**(Suggested maximum time: 5 minutes)**

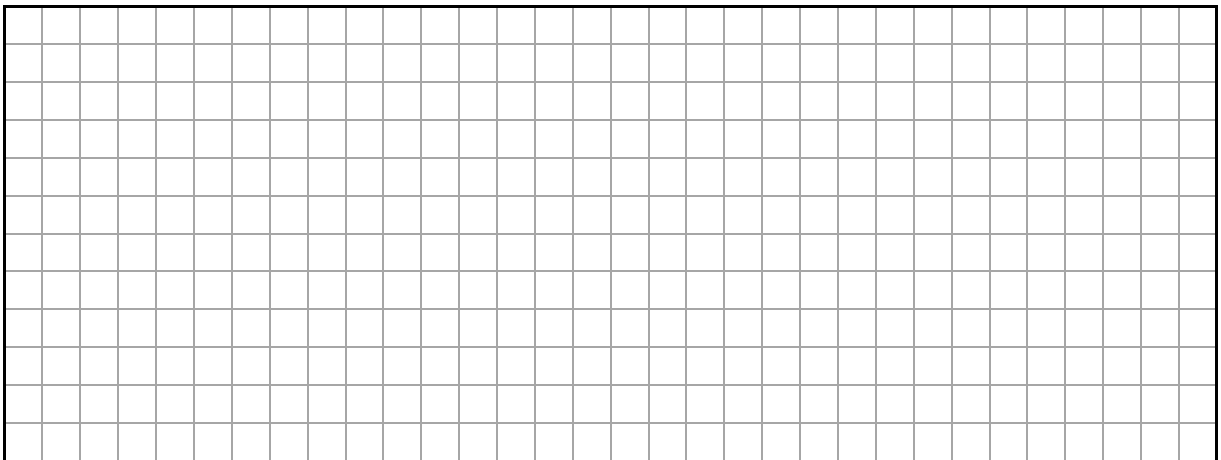
The diagram below shows an equilateral triangle.
The length of each of the sides is given in terms of x .



- (a) Find the value of x and hence work out the length of each side of the triangle.



- (b) Hence work out the length of the perimeter of the triangle.



Question 5 (Suggested maximum time: 10 minutes)

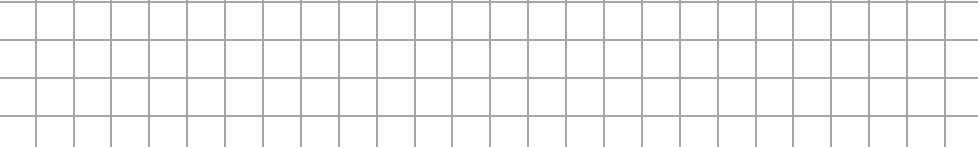
Question 5 (Suggested maximum time: 10 minutes)

- (a)** Factorise fully $25x^2 - 64y^2$.

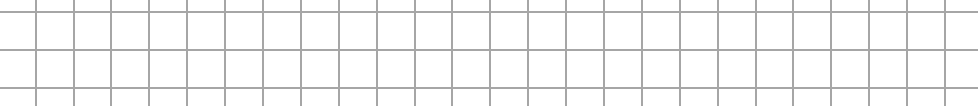
[illegible]

- (b) (i)** Solve the following equation:

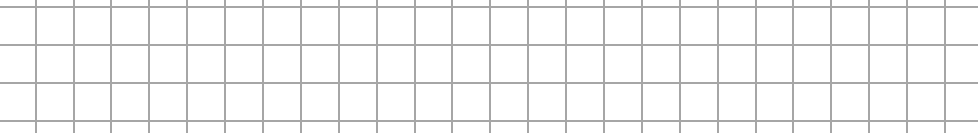
$$\frac{4x+1}{3} - \frac{7x-11}{5} = 3.$$



- (ii) Verify your answer.



- (c)** Simplify $(3x^3 - x^2 - 3x + 1) \div (x - 1)$.



Question 6 (Suggested maximum time: 10 minutes)

Question 6 (Suggested maximum time: 10 minutes)

The number of hours of sunshine in 12 cities throughout the world on a particular day was recorded. The table below shows the results.

6.2	8.3	8.6	8.8	9.1	9.7	10.2	10.5	11.3	11.3	12.2	12.4
-----	-----	-----	-----	-----	-----	------	------	------	------	------	------

- (a)** Explain why the data above is described as numerical data.

[illegible]

- (b)** Data can also be described as categorical. Explain what is meant by categorical data.

[illegible]

- (c) What piece of data from the table above could be referred to as an **outlier**? Give a reason for your answer.

--

Reason:

[illegible]

- (d)** Write down the **modal** number of hours of sunshine recorded (the mode).

- (e) Find the **median** number of hours of sunshine recorded across the 12 cities.

[illegible]

- (f)** Show one method of calculating the **variability** of the data above.

[illegible]

Question 7 (Suggested maximum time: 5 minutes)

Question 7 (Suggested maximum time: 5 minutes)

100 third-year students were asked what sports they had played in the previous week.

The results for the three most popular sports played are as follows:

65 students had played Gaelic football (G)

57 students had played soccer (S)

34 students had played rugby (R)

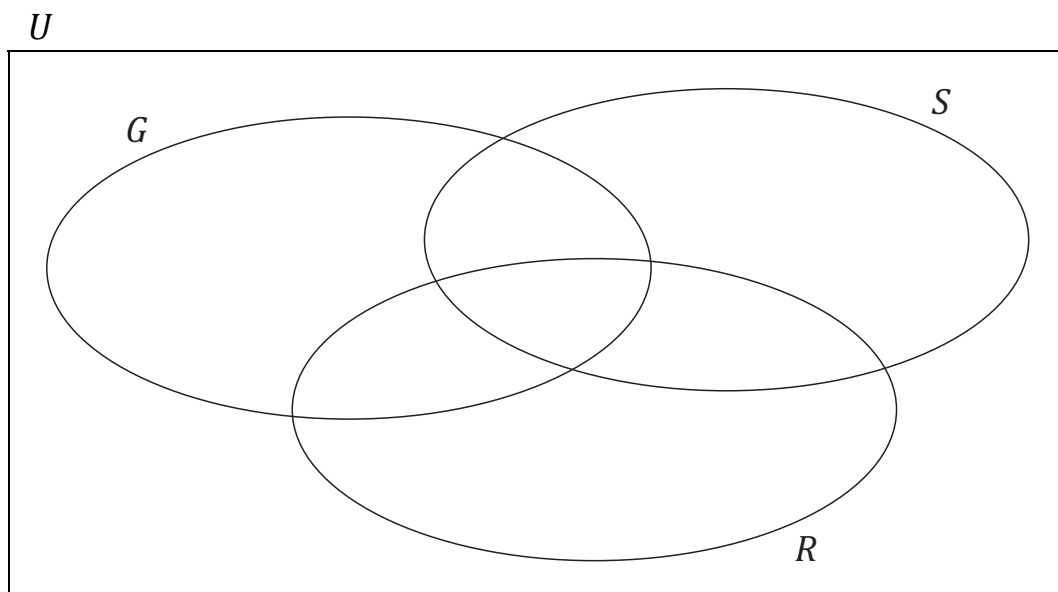
42 students had played Gaelic football and soccer

16 students had played Gaelic football and rugby

x students had played soccer and rugby.

4 students had played all three sports, while 6 students had played none of the sports listed.

- (a)** Use this information to fill in the Venn diagram below, in terms of x .



- (b)** Find x , the number of students who had played soccer and rugby in the previous week.

[illegible]

- (c) Find the **probability** that a student, picked at random, had played just two of the sports listed in the previous week.

[illegible]

Question 8

(Suggested maximum time: 10 minutes)

A women's running group decides to compete in the Boston Marathon. To take part in the race, each person has to run a certified marathon in less than 3 hours and 30 minutes.

Emma's personal best for a marathon to date is 3 hours, 55 minutes.

- (a) Work out the **percentage decrease** Emma needs in her personal best time to qualify for the Boston Marathon.
Give your answer correct to one decimal place.

[illegible]

- (b)** A travel company organises a charter flight from Dublin to Boston for participants and their supporters. It costs €200 000 to charter the flight and it can accommodate 366 passengers. All seats are sold at a cost of €780 each.

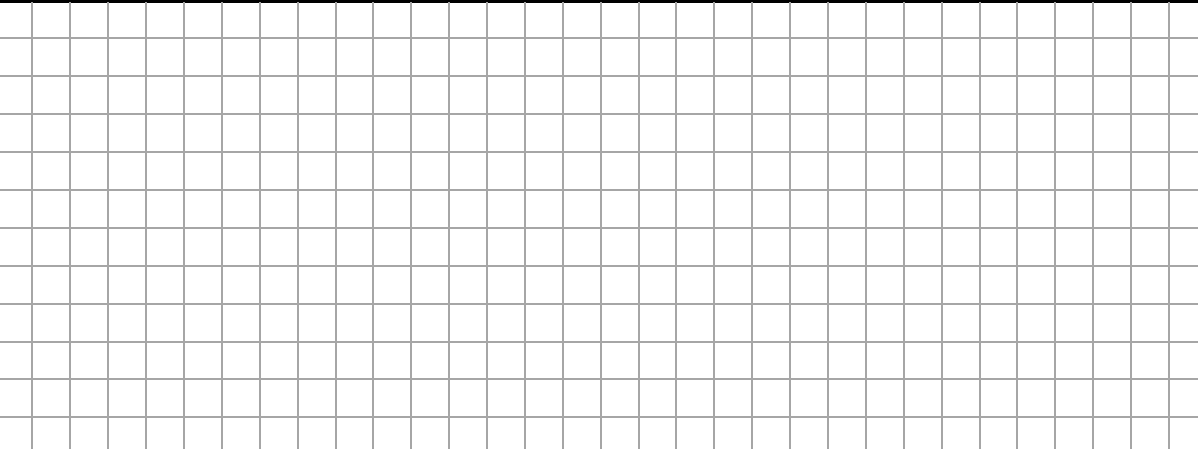
Find the **percentage profit** made by the travel company organising the flight.

Give your answer correct to two decimal places.

[illegible]

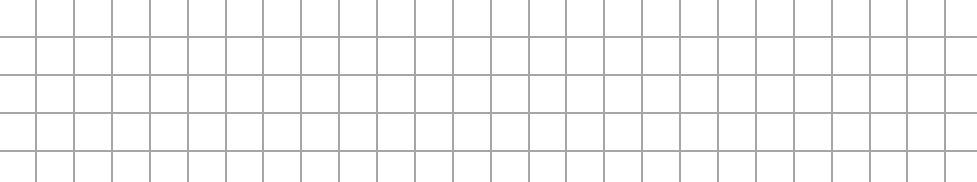
- (c) The group has booked a hotel in Boston for five nights at a cost of \$190 US dollars per room per night. Emma plans to share with two others and the cost of the hotel room will be divided equally between them.

Work out the cost of Emma's hotel accommodation for her stay in Boston in euro (€). The exchange rate is €1 = \$1.20.

A large grid of graph paper, consisting of 20 columns and 10 rows of squares, intended for drawing a picture.

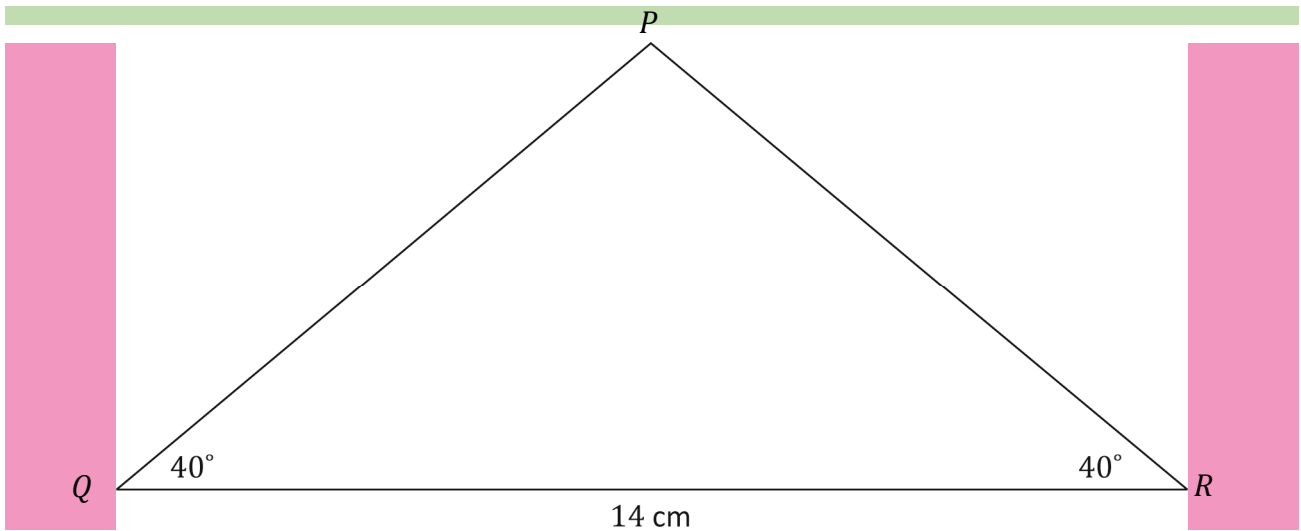
- (d)** Usain Bolt ranks as the fastest runner in the world currently, with a record time of 9.58 seconds for the 100 m sprint.

Work out Usain Bolt's average speed in km/hour.
Give your answer correct to one decimal place.



Question 9**(Suggested maximum time: 15 minutes)**

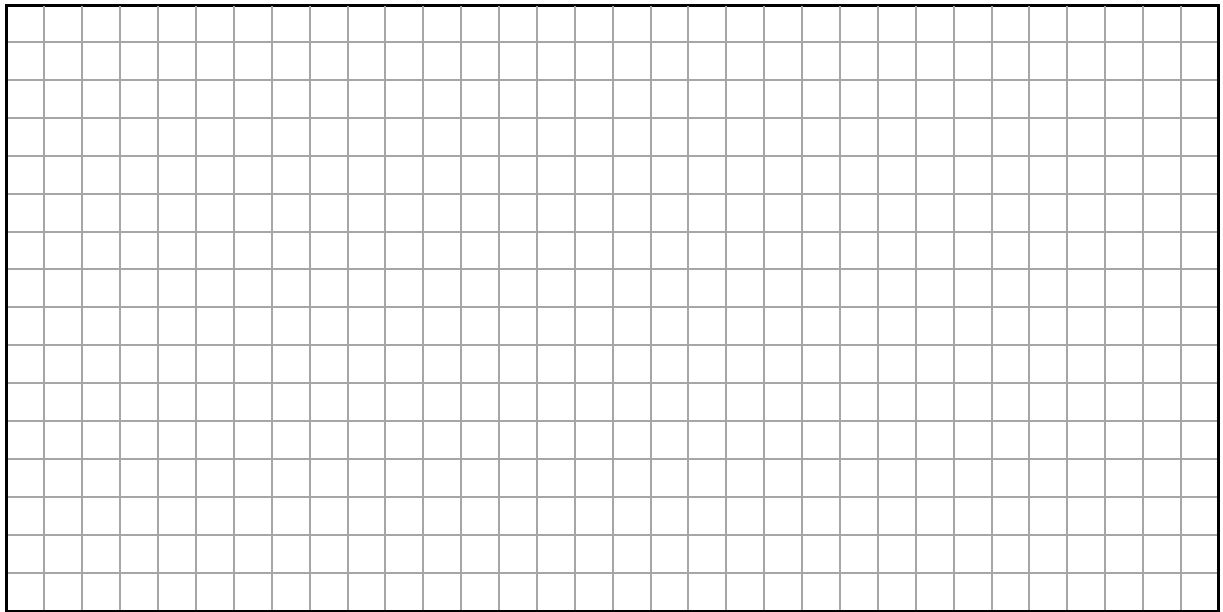
Triangle PQR has a base of length 14 cm and angles of 40° , as shown in the diagram below.



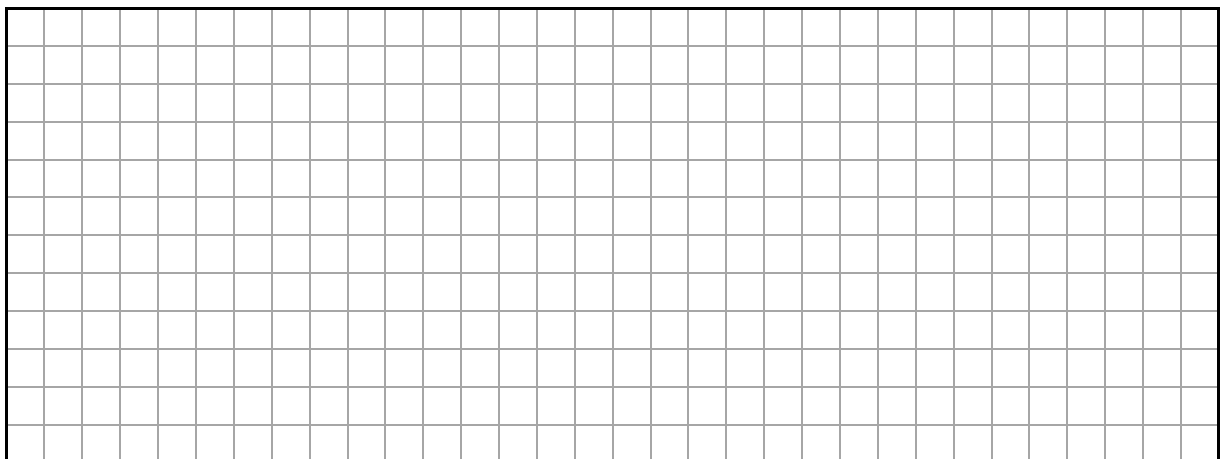
- (a) Without measuring, **construct** a line perpendicular to the line QR passing through the point P . Show any necessary construction lines clearly.

On your diagram label X , the point of intersection of this line and the line QR .

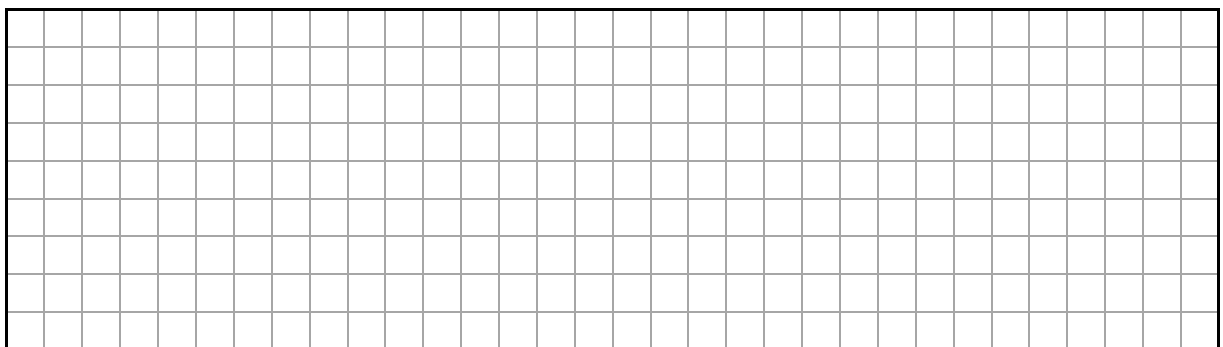
- (b) **Prove** that the triangles PQX and PRX are **congruent**.
Give a reason for each statement that you make in your proof.



- (c) Use **trigonometry** to work out $|PX|$, the perpendicular height of triangle PQR .
Give your answer correct to one decimal place.

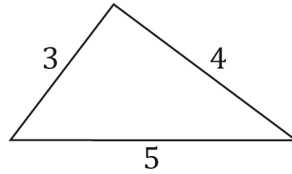


- (d) Work out the area of triangle PQR .

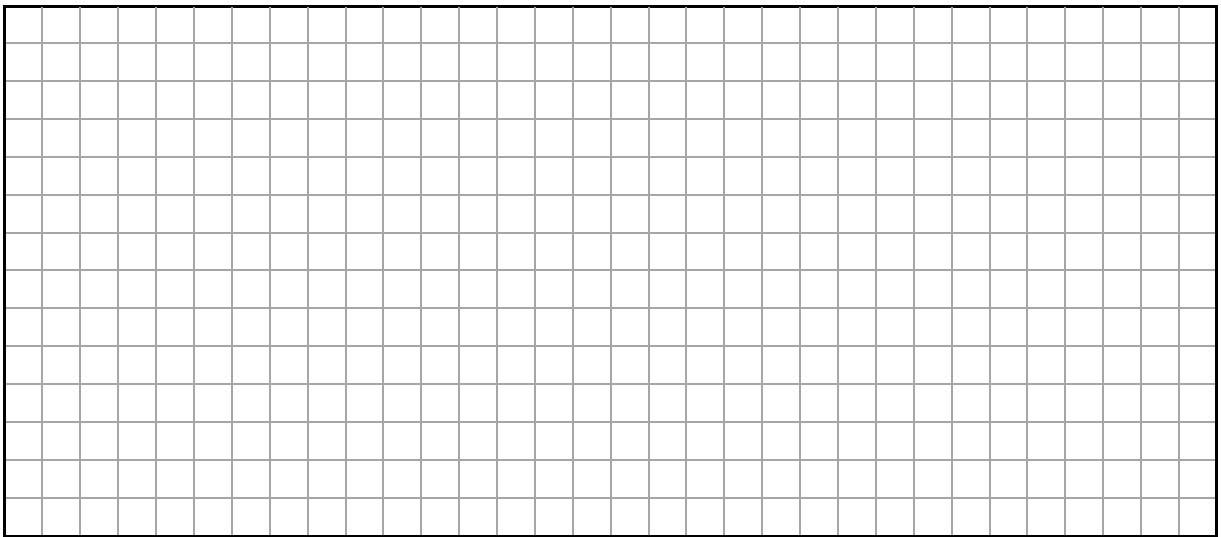


Question 10**(Suggested maximum time: 5 minutes)**

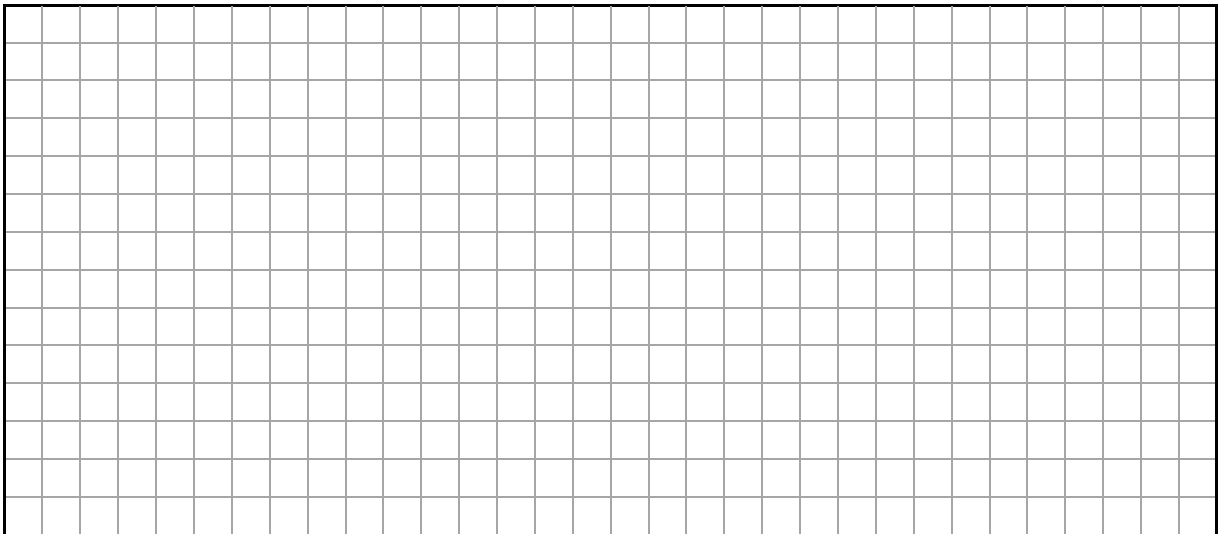
Carpenters and builders regularly use triangles with sides in the ratio 3 : 4 : 5 to make sure they are constructing a precise right angle.



- (a) Indicate the right angle in the diagram above.
Ensure that it is clearly labelled.
- (b) Use a **theorem** from your course to prove that triangles with sides in the ratio 3 : 4 : 5 have a right angle.



- (c) Use **trigonometry** to work out the size of the smallest angle in the diagram above.
Give your answer correct to the nearest degree.



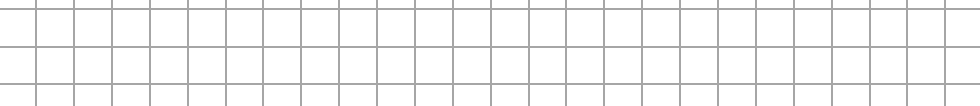
Question 11 (Suggested maximum time: 15 minutes)

Question 11 (Suggested maximum time: 15 minutes)

- (a)** The first five terms of a linear pattern are shown below.

$-7, \quad -2, \quad 3, \quad 8, \quad 13.$

- (i) Find the sum of the first 8 terms of this pattern.



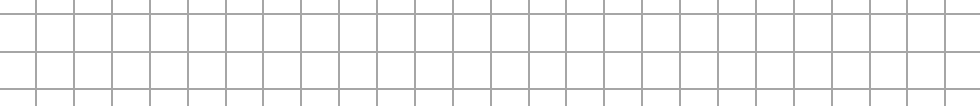
- (ii) Describe in your own words how consecutive terms of this pattern are formed.



- (iii) Hence, or otherwise, write an algebraic expression to represent the **n th term** of this pattern.



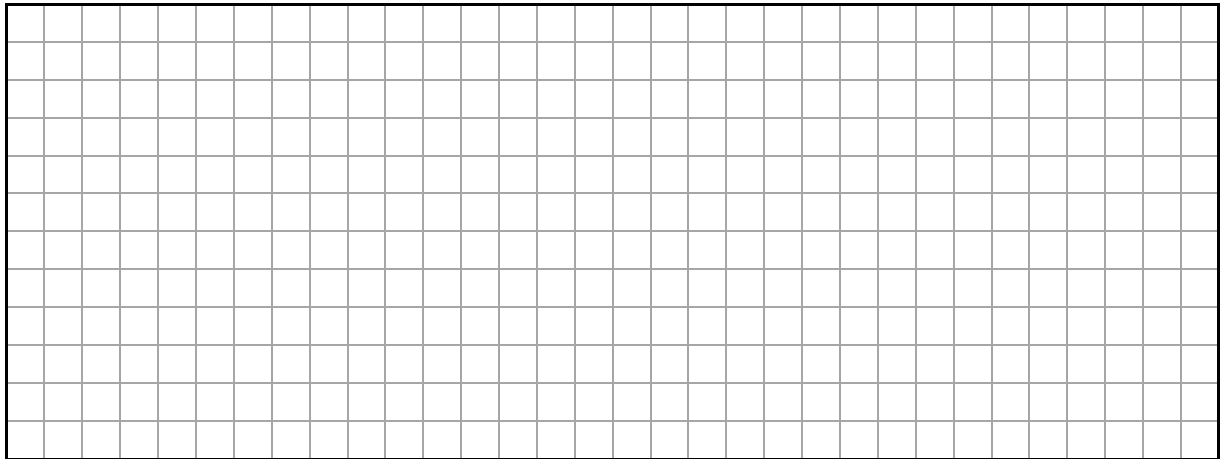
- (iv) Work out the **50th term** of this pattern.



- (b) The first five terms of a **quadratic** pattern are shown below.

Term	1	2	3	4	5
Value	5	5	7	11	17

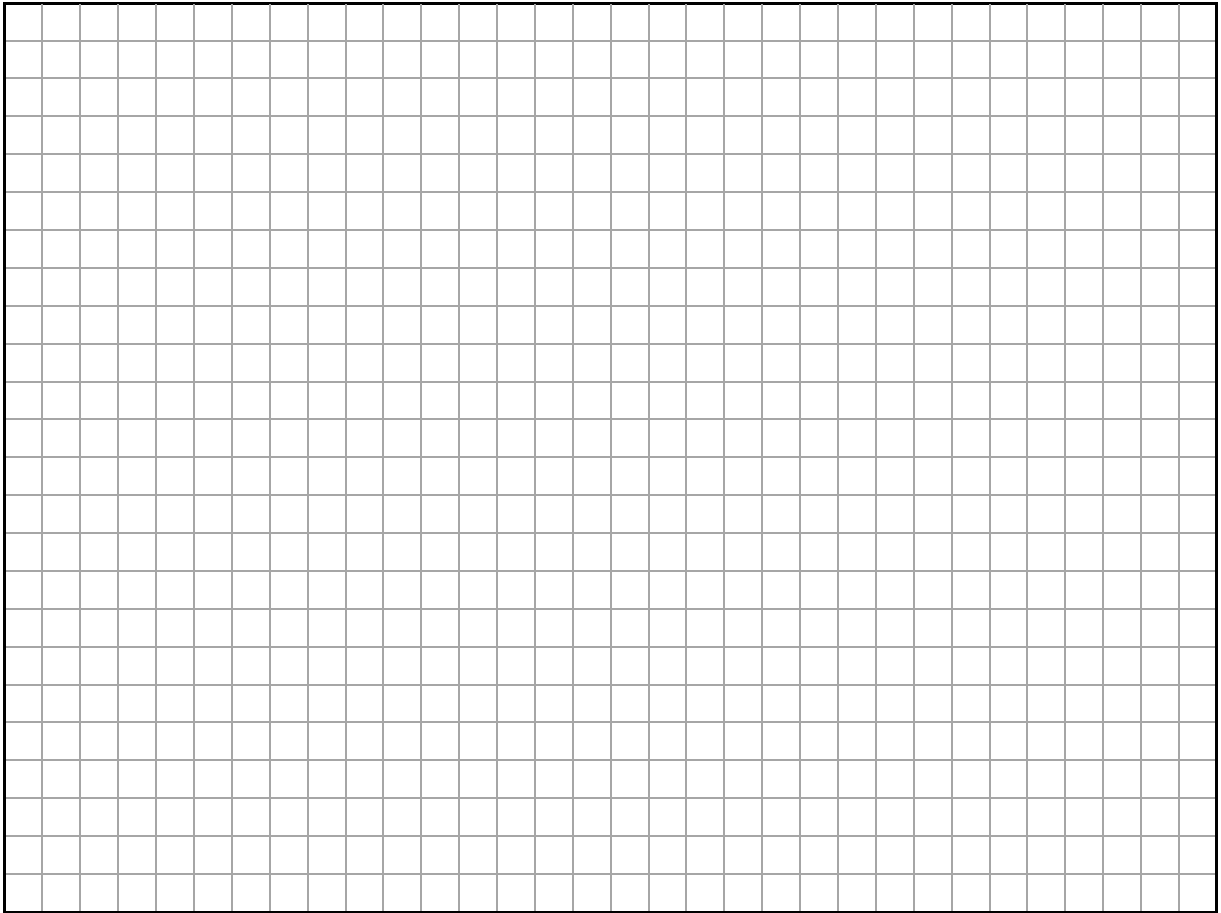
- (i) Show that this is a quadratic pattern.



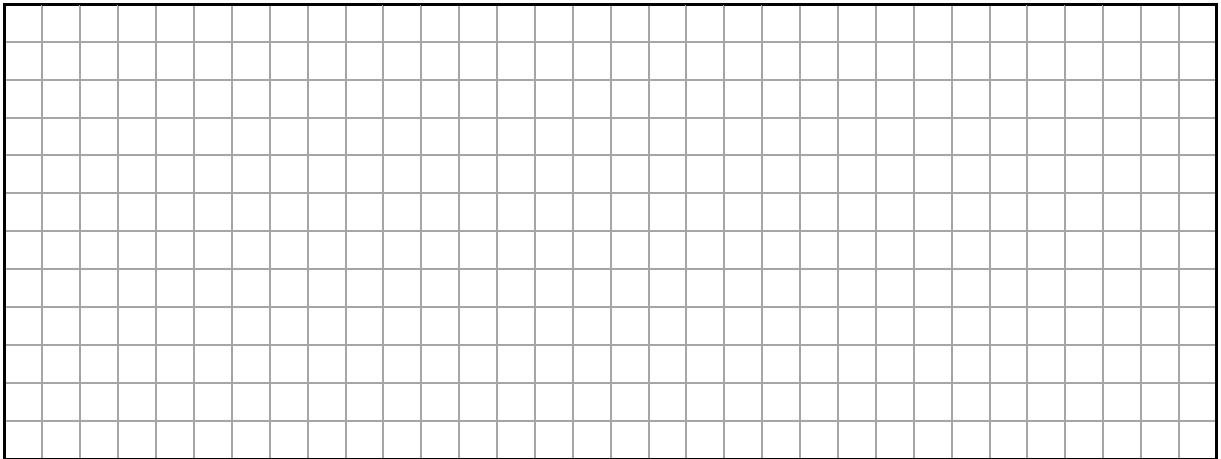
- (ii) Draw the graph of the first five terms of the pattern by putting the term number on the horizontal axis and the value on the vertical axis.



(iii) Write an algebraic expression to represent the n th term of this pattern.



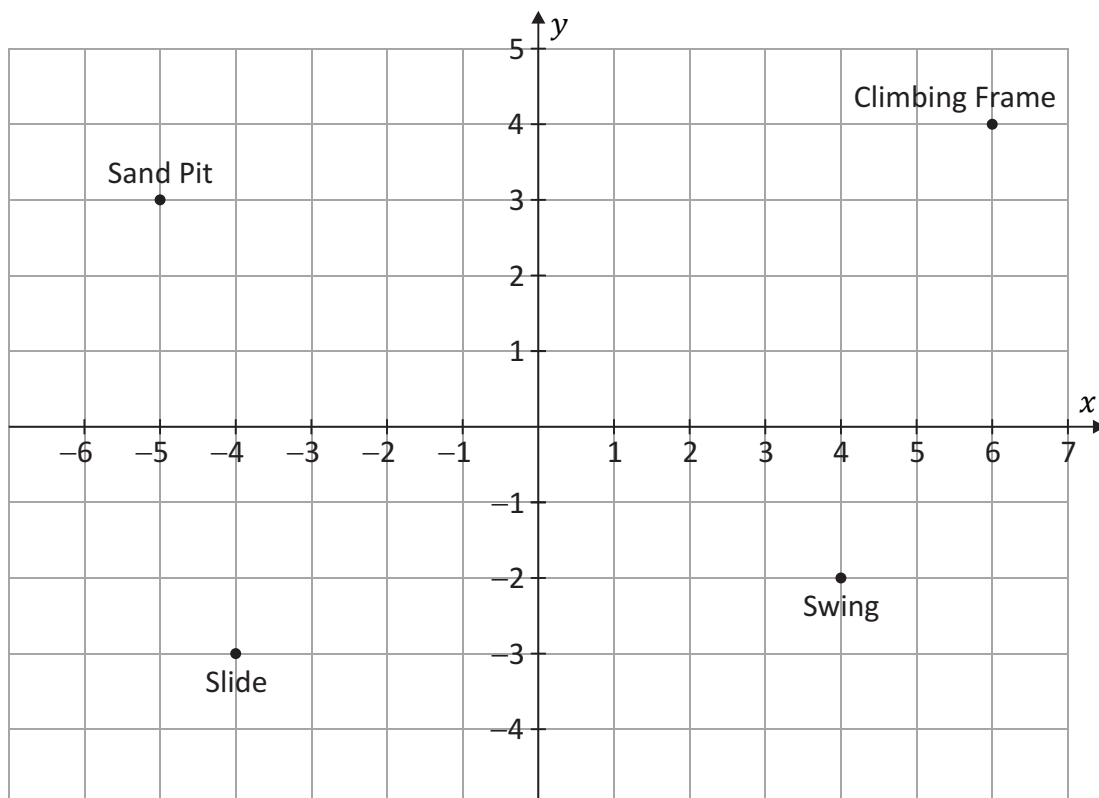
(iv) What term number of this pattern has a value of 77?



Question 12

(Suggested maximum time: 10 minutes)

The co-ordinated map below shows the locations of the climbing frame, sand pit, slide and swing in a playground. Each square on the map is 1 cm in length.



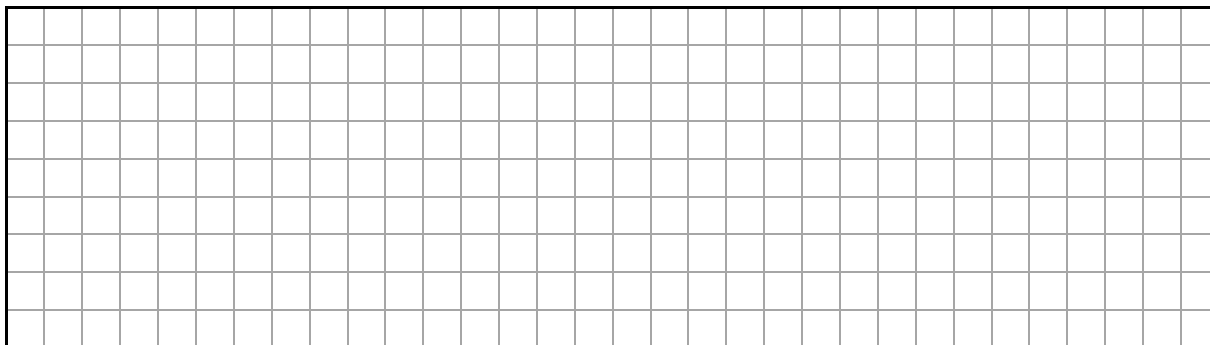
- (a) Write down the co-ordinates of the climbing frame, sand pit, slide and swing in the table below.

Item	Climbing Frame	Sand Pit	Slide	Swing
Co-ordinates	(,)	(,)	(,)	(,)

- (b) A path goes directly from the sand pit to the swing. The slope of the path is $-\frac{5}{9}$.

Work out the equation of a line that represents this path.

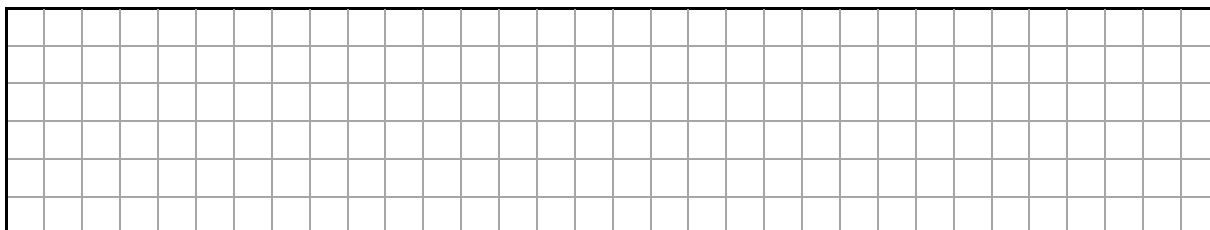
Give your answer in the form $ax + by + c = 0$, where a , b and $c \in \mathbb{Z}$.



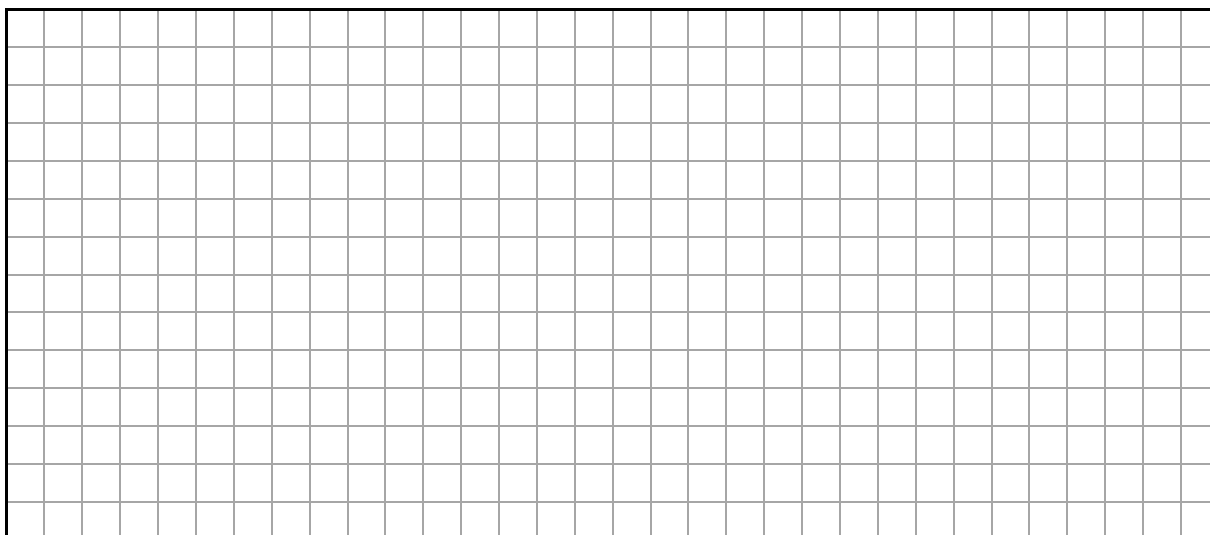
- (c) A seesaw is added to the playground. Its position on the map is the image of the position of the climbing frame under **central symmetry** in the origin (0, 0).
Write down the co-ordinates of the seesaw and indicate its position on the map above.

Co-ordinates of seesaw =

(,)



- (d) Work out the **distance** from the slide to the seesaw on the diagram.
Give your answer in cm, correct to two decimal places. Show your working out.



- (e) The map is drawn to a scale of 1:300.
It is recommended that there should be at least 4 metres between each piece of playground equipment.
Does the position of the seesaw follow this recommendation? Justify your answer.

Answer:	
Justification:	

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Copyright notice

This examination paper may contain text or images for which DEB Exams is not the copyright owner, and which may have been adapted, for the purpose of assessment, without the authors' prior consent. This examination paper has been prepared in accordance with Section 53(5) of the *Copyright and Related Rights Act, 2000*. Any subsequent use for a purpose other than the intended purpose is not authorised. DEB Exams does not accept liability for any infringement of third-party rights arising from unauthorised distribution or use of this examination paper.

Pre-Junior Cycle Final Examination, 2022

Mathematics – Higher Level

Time: 2 hours

4e6b4069-6072-49fb-a557-f5acf7e96109



© 2022-J020-1-EL-20/20